

Modeling Kilonova in Python

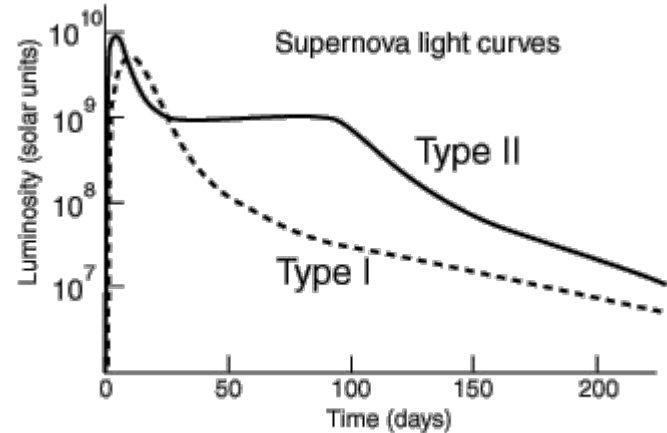
TIMESTEP Advanced Python

Astronomical Transient

A transient is an astrophysical event that changes in brightness on human timescales

Examples include:

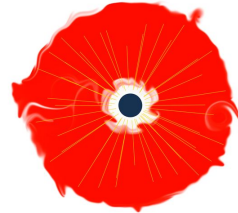
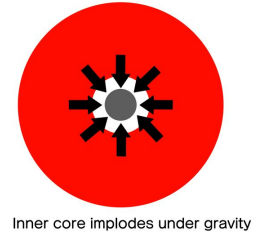
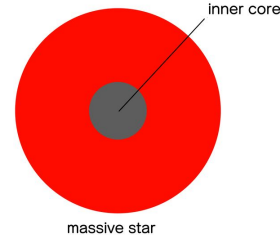
- Supernova
- Gamma-ray burst
- Tidal disruption event
- etc.



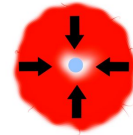
Adapted from Chaisson & McMillan

Neutron Stars

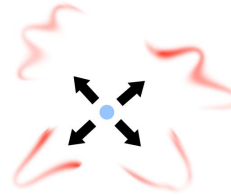
When a massive star dies, it leaves behind a dense core of neutrons, called a neutron star



Gravity smashes electrons and protons together, forming neutrons, and releasing a shower of neutrinos. Outer layers slosh violently from standing accretion shock instability.



Outer layers implode and collapse onto the inner core at 25% the speed of light.



Outer layers bounce off the dense core, creating a supernova.

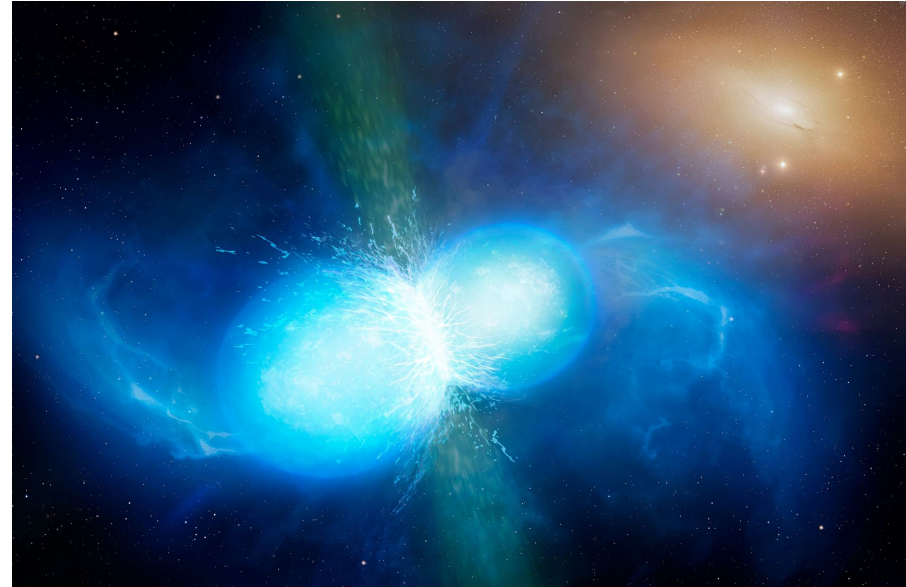


The resultant free core is a neutron star.

Binary Neutron Star Merger

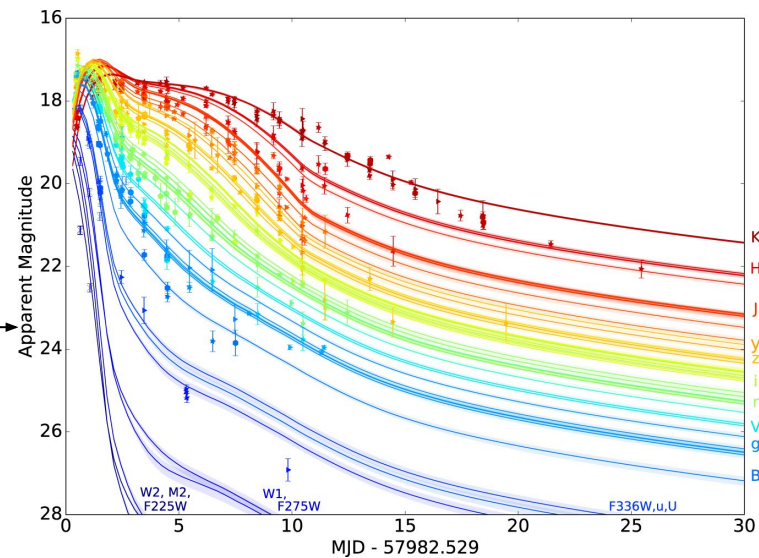
When two neutron stars merge

This is visible as both gravitational waves and as an electromagnetic *transient*



Kilonova

The *transient event* that is visible in the optical wavelengths following a binary neutron star merger

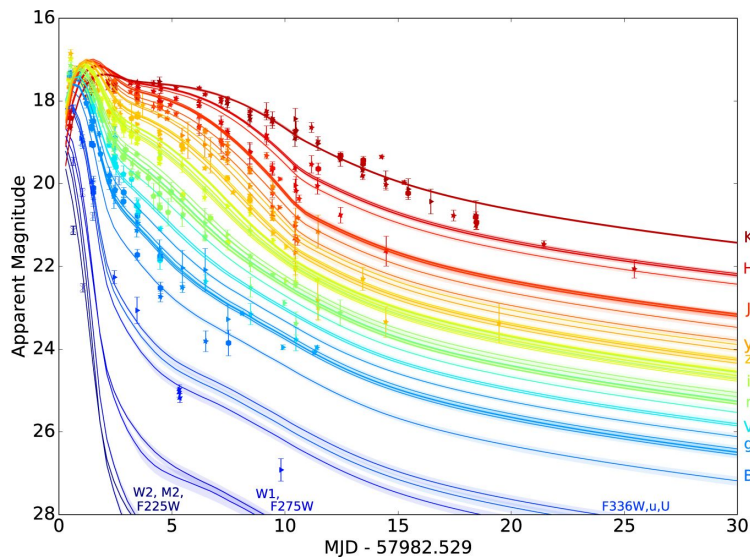


Villar+2017

GW170817 & AT2017gfo

The binary neutron star merger called GW170817 was discovered in 2017 by the LIGO and Virgo gravitational wave detectors

AT 2017gfo is the Kilonova associated with this binary neutron star merger, and is the only optical counterpart to a binary neutron star merger known!



Villar+2017

The Kasen+17 Model Grid

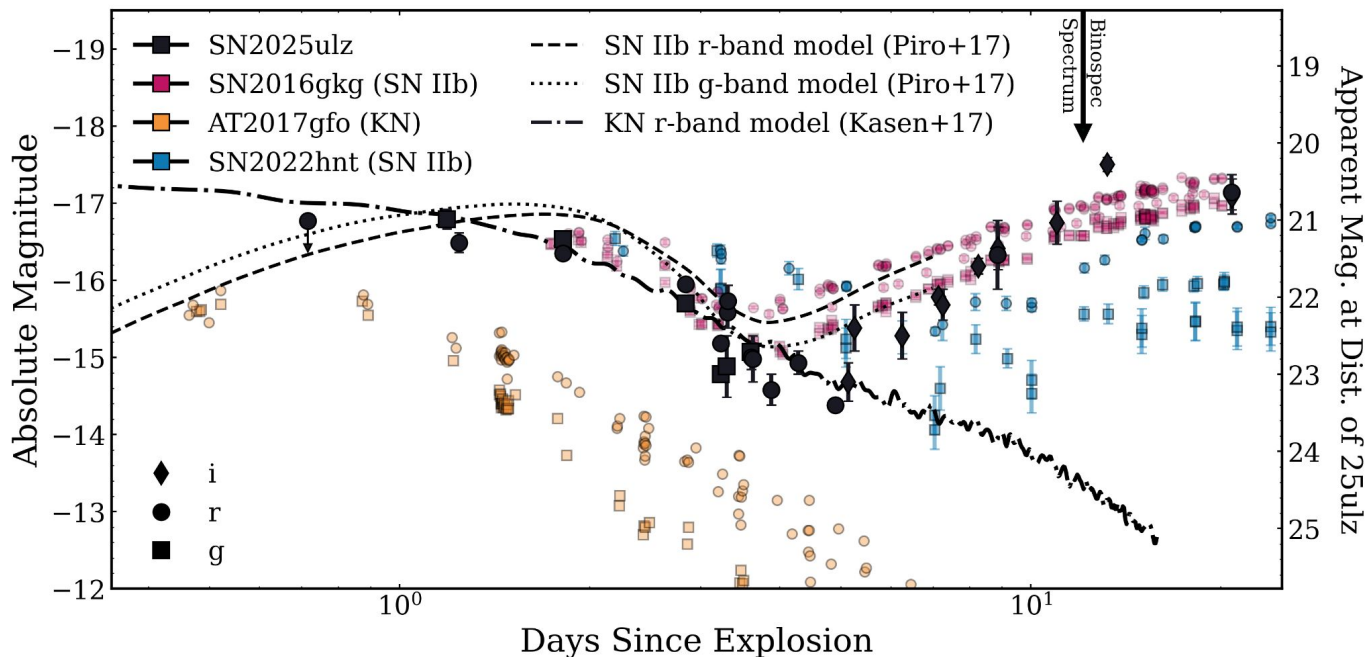
In 2017, and soon after GW170817, Kasen et al. came out with a “grid” of Kilonova models

These models have the following parameters:

- Ejecta Mass
- Ejecta Velocity
- Lanthanide Fraction

Your Mission

Figure out which model in the Kasen+17 grid fits the r-band light curve the best!



Pairs for Working on the Notebook

Justin & Riley

Luke & Brianna

Emma & Brandon

Sydney & Sanjana

Ben & Isabelle

Grecia & Kate

Hints

Part 1

```
at2017gfo_table = read_csv("AT2017gfo-lightcurve.csv")  
print(at2017gfo_table.columns)
```

Part 2, Cell 1

```
mag_column = at2017gfo_table["mag"]
```

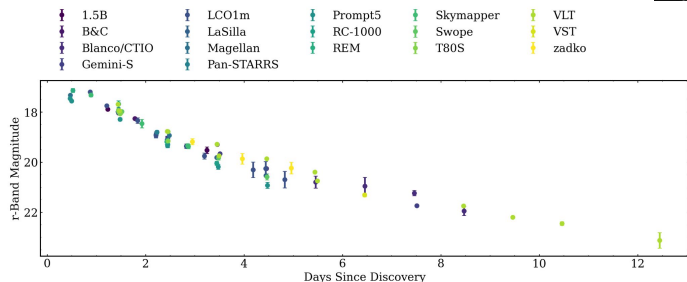
```
mag_column
```

Part 2, Cell 2

```
data = at2017gfo_table[at2017gfo_table["Band"] == "r"]
```

```
data
```

Part 2, Cell 3



```
: fig, ax = plt.subplots(figsize=(14,6))

teles = np.unique(data["Telescope"])
cmap = plt.cm.get_cmap("viridis")
colors = cmap(np.linspace(0,1,len(teles)))
for tele, color in zip(teles, colors):
    grp = data[data["Telescope"] == tele]
    ax.errorbar(
        grp["Phase"],
        grp["mag"],
        yerr=grp["e_mag"],
        linestyle = "none",
        marker = "o",
        capsize = 3,
        label = tele,
        color = color
    )
```

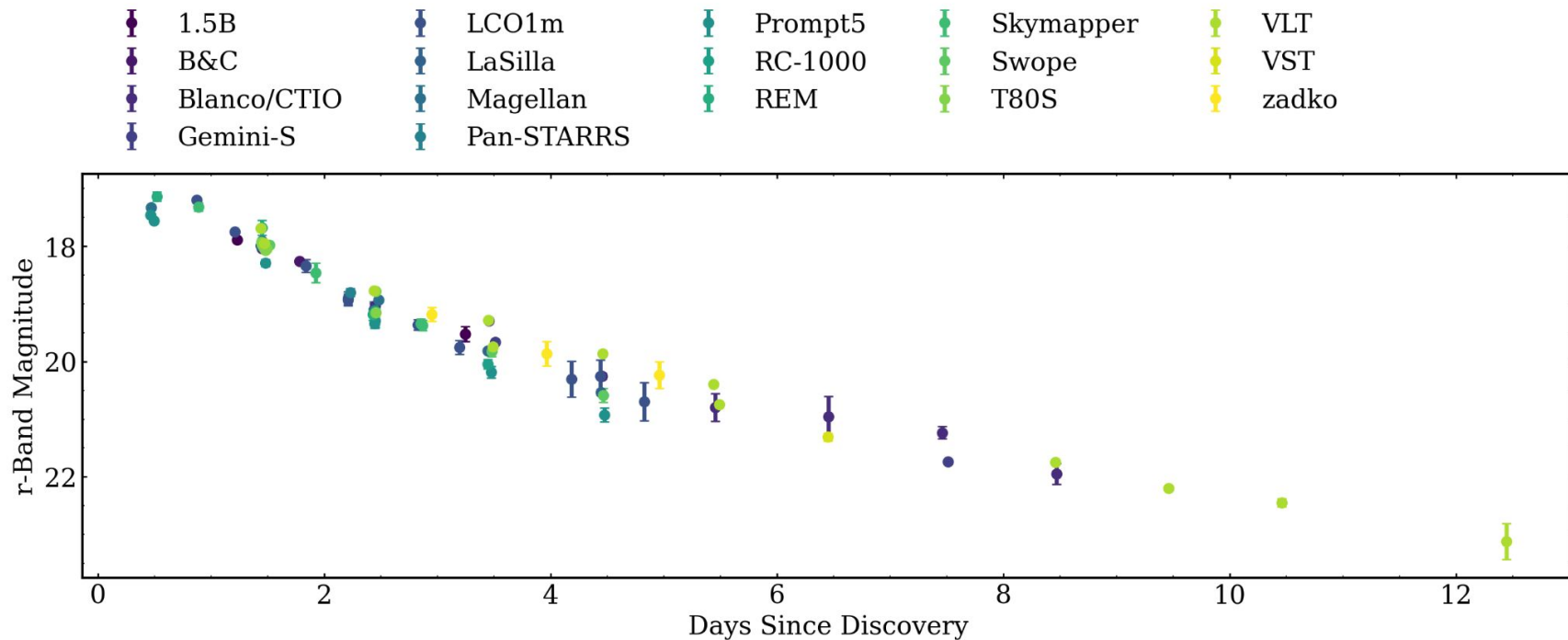
```
ax.invert_yaxis()
```

```
ax.set_ylabel("r-Band Magnitude")
```

```
ax.set_xlabel("Days Since Discovery")
```

```
ax.legend(ncols=5, loc="lower left", bbox_to_anchor=(0,1))
```

Part 2, Cell 3 Plot



Part 3

```
: model = read_kasen17_model("kasen_kilonova_models_2017/knova_d1_n10_m0.020_vk0.10_Xlan1e-3.0.h5")  
model
```

Part 4

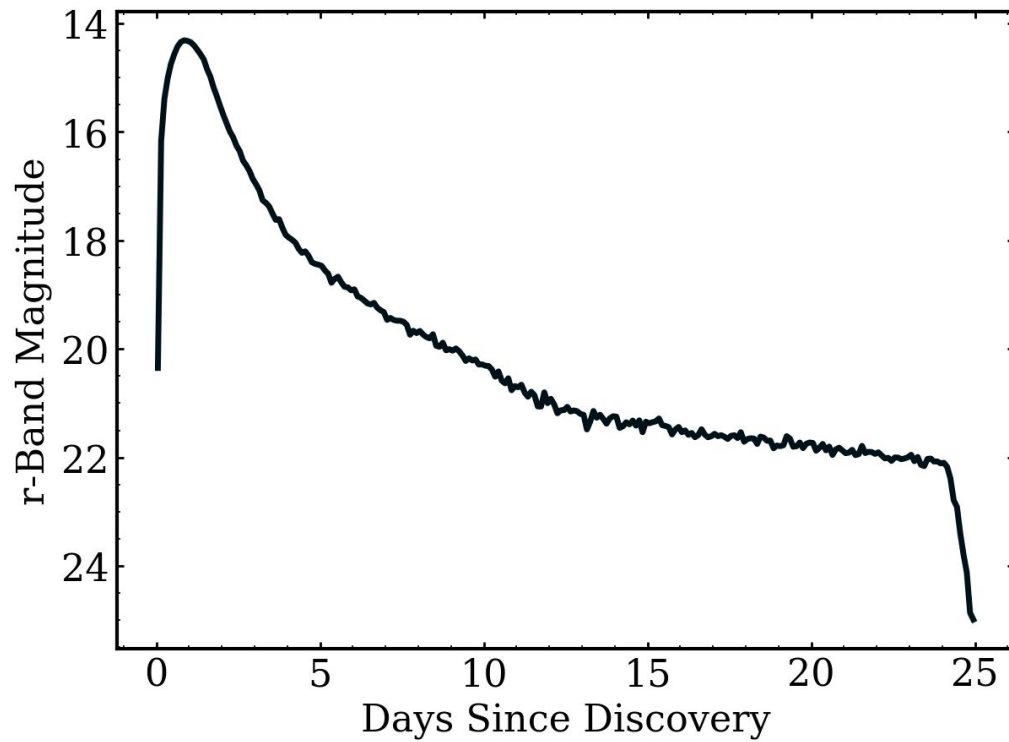
```
fig, ax = plt.subplots()

ax.plot(
    model["dt"],
    model["mag"],
    linestyle = "-",
    marker = "none",
)

ax.invert_yaxis()

ax.set_ylabel("r-Band Magnitude")
ax.set_xlabel("Days Since Discovery")
```

Part 4 Plot



Part 5

```
def sum_square_residual(data:np.ndarray, model:np.ndarray) -> np.ndarray:  
    return np.sum((data-model)**2)
```

Part 6

```
lst_sq = np.inf
bestfitfile = ""

for file in glob.glob("kasen_kilonova_models_2017/*h5"):
    model = read_kasen17_model(file)

    model = model[np.isfinite(model["mag"])]

    # find the rows corresponding to each Phase in the data
    dt_idx = np.array([
        np.argmin(
            np.abs(
                t-model["dt"]
            )
        ) for t in data["Phase"]
    ])

    residual = sum_square_residual(data["mag"], model[dt_idx]["mag"])
    if residual < lst_sq:
        lst_sq = residual
        bestfitfile = file

bestfitfile
```

Part 7

```
model = read_kasen17_model(bestfitfile)
model = model[np.isfinite(model["mag"])]

fig, ax = plt.subplots(figsize=(14,6))

for tele in np.unique(data["Telescope"]):
    grp = data[data["Telescope"] == tele]
    ax.errorbar(
        grp["Phase"],
        grp["mag"],
        yerr=grp["e_mag"],
        linestyle = "none",
        marker = "o",
        capsize = 3,
        label = tele
    )

ax.plot(
    model["dt"],
    model["mag"],
    linestyle = "-",
    marker = "none",
)

ax.invert_yaxis()
```

Part 7 Plot

